

When Fairness Metrics Disagree Under Shift: A Simulation Study of Metric Distortion and Mitigation Trade-offs

A stylized case study of a medical follow-up risk score under hospital deployment shift

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Motivation & High-Level Idea

Group fairness metrics such as demographic parity and equalized odds are usually defined under a fixed data-generating process. However, real-world deployments of clinical decision support systems rarely enjoy such stability: patient populations differ across hospitals, covariate distributions drift over time, and label noise (e.g., mis-coded diagnoses) may become asymmetric across demographic groups.

In this project, we study a simplified setting inspired by a binary risk score that decides whether a patient should be referred for further testing / specialist follow-up. A logistic regression model is trained on data from a “source” hospital and then evaluated under controlled distribution shifts that mimic deploying the same model in a different “target” hospital with different patient mix and coding patterns.

We ask:

- How do standard group fairness metrics behave under controlled distribution shifts, even when the underlying classifier remains fixed?
- Under what conditions can demographic parity remain stable or appear comparatively favorable while error-based metrics degrade?
- How do importance reweighting, post-hoc threshold adjustment, and target-distribution re-training affect the observed fairness distortion, and what trade-offs do they incur?

Theoretical Framing

We consider binary classification with two patient groups A, B (e.g., defined by a protected attribute such as race or sex). A fixed logistic regression model outputs a risk score used to decide whether to recommend a follow-up test / specialist referral. The model is trained on an initial “source hospital” distribution.

Here, the label represents a recorded proxy for need of follow-up; under asymmetric label noise, this proxy may be corrupted differently across groups.

We track four group fairness metrics for this referral decision: demographic parity, equalized odds, calibration error, and the true-positive-rate gap (TPR gap) between groups.

We consider three shift scenarios, each of which can be interpreted as a stylized hospital deployment shift, with a severity parameter:

Shift type	Severity	Definition	Clinical intuition (informal)
Group prior shift	$\alpha \in [0, 1]$	$P(Y = 1 A) = 0.3 + \alpha \cdot 0.2$ $P(Y = 1 B) = 0.3 - \alpha \cdot 0.2$	Target hospital has a higher disease prevalence in group A than B
Covariate shift	$\gamma \in [1, 4]$	Multiply standard deviation / scale of feature X_1 for group A by γ	Target hospital sees a different distribution of a key risk factor
Asymmetric label noise	$\beta \in [0, 0.3]$	Flip positive labels in A with prob β ; negative labels in B with prob $\beta/2$	Diagnosis / coding errors are asymmetric across groups

Empirical Setup

- **Data:** Synthetic binary classification with two patient groups, each with 2-dimensional features (one informative “clinical” feature, one noisy feature). The outcome label is a recorded proxy for whether the patient needs follow-up.
- **Experiments:** For each shift scenario, we sweep the severity parameter and compute all fairness metrics on test sets. The main experiment keeps the referral model fixed (trained once on unshifted “source hospital” data) to isolate the effect of deployment shift on metric behavior.
- **Mitigation Attempts:** Evaluate three strategies to detect or correct fairness distortion under shift:
 1. **KDE Importance Reweighting:** Estimate density ratios via kernel density estimation and reweight the training loss to better match the target-hospital distribution.
 2. **Post-hoc Threshold Adjustment:** Tune group-specific decision thresholds to equalize TPR across groups (Equal Opportunity) while tracking the impact on other metrics.
 3. **Target Retraining:** Retrain the referral model directly on samples from the shifted distribution (as if new labeled data from the target hospital were available).
- **Output:** Phase diagrams (metric value vs. severity, separate panels per shift type), identification of “danger zones”, comparison of different mitigation methods and open-source simulation code.

References

- [1] Asiaee, A. & Aryan, K. (2026). Fairness under group-conditional prior probability shift: Invariance, drift, and target-aware post-processing. *arXiv preprint*, arXiv:2602.05144.
- [2] Chen, Y., Raab, R., Wang, J., & Liu, Y. (2022). Fairness transferability subject to bounded distribution shift. *arXiv preprint*, arXiv:2206.00129.
- [3] Barrainkua, A., Gordaliza, P., Lozano, J. A., & Quadrianto, N. (2022). A survey on preserving fairness guarantees in changing environments. *arXiv preprint*, arXiv:2211.07530.
- [4] Shao, M., Li, D., Zhao, C., Wu, X., Lin, Y., & Tian, Q. (2024). Supervised algorithmic fairness in distribution shifts: A survey. *arXiv preprint*, arXiv:2402.01327.